

Solomon Ahedor

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EDUCATION

Kwame Nkrumah University of Science and Technology (KNUST) | Kumasi, Ghana

Bachelor of Science, Metallurgical Engineering | Expected Completion: 2026

- **Relevant Coursework:** Extractive Metallurgy, Mineral Processing, Data Science, Phase Transformation, Materials Characterization, Machine Learning, Artificial Intelligence, Data Science.

TECHNICAL SKILLS

- **Metallurgical Processes:** Refractory Ore Processing, Bacterial Oxidation (BIOX), Froth Flotation, Carbon-in-Leach (CIL), Elution Circuits.
- **Laboratory Instrumentation:** LECO (S832 DR/S632) for Sulfur and Carbon analysis, Atomic Absorption Spectroscopy (AAS), Particle Size Analyzers, Optical Microscopy.
- **Computational Modeling & Data:** Python (Pandas, NumPy, Matplotlib), do-mpc (Model Predictive Control), Machine Learning (Scikit-learn), MATLAB.
- **Software & Dev:** Microsoft Office Suite, Python, Design Expert, MATLAB.

PROFESSIONAL EXPERIENCE

AngloGold Ashanti (Obuasi Mine), Obuasi, Ghana

Metallurgy & Process Engineering Intern, Nov 2025 – Dec 2025

- **Plant Operations:** Completed technical rotation through the full refractory gold processing circuit including Crushing, Milling, Flash and Inline Leach Reactor (ILR), Flotation, BIOX and CCD, and CIL operations.
- **Process Control:** Monitored BIOX bacterial activity and performed ferrous/ferric ion determinations to support optimization of sulfide oxidation rates and CIL adsorption efficiency.

- **Laboratory Analysis:** Performed extended flotation tests and Particle Size Analysis (PSA) in the Met-lab to evaluate recovery efficiency and grind kinetics, providing data to refine process settings.
- **Analytical Chemistry:** Assisted with LECO instrumentation (S832 DR and S632) for total sulfur, carbon, and carbonate analysis, and utilized AAS for detecting gold and arsenic concentrations in solution.
- **Reporting:** Delivered a technical presentation to management on operational workflows and process-monitoring recommendations for CIL and Elution, supporting decision-making on site practices.

RESEARCH & ACADEMIC PROJECTS

Department of Materials & Metallurgical Engineering, KNUST, Kumasi, Ghana Undergraduate

Researcher – Process Dynamics & Optimization, Sept 2025 – Present

- **Project:** Optimizing Froth Flotation Performance through Dynamic Modelling.
- **Modelling:** Trained an XGBoost models to predict grade and recovery on small and large datasets, achieving R^2 scores of 65% to 95%.
- **Control Systems:** Implemented NMPC and MHE to handle process non-linearities such as parabolic air recovery and reduce model degradation.
- **Simulation:** Using Python libraries (Pandas, NumPy) and the do-mpc toolbox to simulate control architectures for improving grade and recovery targets.

AGRAVOLUTION (Eco-Friendly Silica/Glass Project), KNUST

Research and Development Lead, 2023 – Present

- **Leadership:** Led a team of 3 undergraduate researchers to synthesize silica/glass compositions from agricultural waste, developing a low-carbon alternative to conventional silicate glass and enriched silica.
- **Experimentation:** Designed experimental protocols to characterize material properties using microscopy and thermal analysis, generating data to correlate microstructure with performance.
- **Mentorship:** Coordinated laboratory schedules and trained peers in proper safety and experimental techniques.

Bio-based & Biodegradable plastic film and molded container, KNUST - MIH

Project Lead

- **Innovation:** Engineering a biodegradable plastic from cassava starch
- **Analysis:** Optimizing water adsorption kinetics, yield strength and other synthesis parameters to maximize the mechanical properties.
- **Impact:** Initiated the project to address environmental pollution and presented it in university innovation competitions.

MetaForge Assistant App, KNUST

Project Designer & Developer

- Developed a computational tool for metallurgy students featuring phase diagram visualization, casting parameter optimization, and fatigue life simulation, improving access to core materials-engineering calculations and analysis.

LEADERSHIP & AWARDS

- **President**, Materials Innovation Hub (MIH), KNUST, *2026 Term*
Elected to support student innovation and materials science projects.
- **Citation Recipient**, Hult Prize National Competition, *2025*
- **2nd Runner-Up**, Hult Prize KNUST Campus Edition, *2025*
- **Zonal Participant**, Mastercard-Kosmos Innovation Center Competition, *Awarded \$300 for Market Research*

LANGUAGES

- English (Fluent)
- Twi (Fluent)
- German (Basic)